HW-02 — Manual Event Simulation

Module M03 | Chapters 3–5 | Weight 5% | Difficulty ⋆⋆

Module: M03 Chapters: 3–5 Weight: 5% Due: End of Week 4

Overview

Manually trace 10 customers through a single-server pharmacy queue and compute all standard performance measures from first principles. No SimPy — this assignment teaches what SimPy does under the hood.

1 System Description

A pharmacy has one dispensing counter (server). Customers arrive and queue in FCFS order. Use the arrival and service times in Table 1.

Customer	C1	C2	C3	C4	C5	C6	C7	C8	C9	C10
Arrival time (min)	0	3	8	11	17	22	25	29	34	38
Service time (min)	4	7	3	6	2	5	4	3	6	4

Table 1: Pharmacy queue — given arrival and service times.

2 Parts

2.1 Part A — Event Trace Table (25 pts)

Build a complete event trace table in chronological order. For every event (arrival or departure) record: time, event type, customer number, queue length $Q(t)$ *before* the event, server status (idle / busy / transition), cumulative area under $Q(t)$.

The trace must contain exactly **20 events** (10 arrivals + 10 departures). Note: at $t = 34$, customer C9 arrives at the same moment C8 departs — state the tie-breaking convention you use.

2.2 Part B — Performance Measures (35 pts)

From the completed trace, compute:

1. Individual waiting times $W_q^{(i)}$ for each customer (time from arrival to service start)
2. $\bar{W}_q$ — sample mean waiting time in queue

3. $\bar{W}$ — sample mean sojourn time (arrival to departure)
4. $\hat{\rho}$ — observed server utilization
5. $\bar{L}_q$ — time-average queue length (from area under $Q(t)$)
6. $\bar{L}$ — time-average number in system

Show all arithmetic.

2.3 Part C — Little's Law Verification (20 pts)

Use your results to verify both forms of Little's Law:

$$\bar{L} \approx \hat{\lambda}\,\bar{W} \qquad \text{and} \qquad \bar{L}_q \approx \hat{\lambda}\,\bar{W}_q$$

where $\hat{\lambda} = 10/T_{\text{sim}}$. Report the relative error for each equation. Discuss why the error is not zero.

2.4 Part D — Python Confirmation Script (20 pts)

Write a Python script (no SimPy — use plain lists and arithmetic) that:

- Reads the given arrival and service times
- Simulates the queue event-by-event
- Prints $\bar{W}_q$, $\bar{W}$, $\hat{\rho}$, $\bar{L}_q$, $\bar{L}$

Your printed output must match your hand calculations to within rounding.

3 Format and Submission

- **Written report** (PDF, ≤ 5 pages): Parts A–C with full trace table
- `hw02_trace.py`: Part D script
- Submit via GitHub Classroom; file names: `hw02_`*`netid`*`.pdf` and `hw02_trace.py`

Grading Summary

Component	Points
Part A — Event trace (20 events, correct order)	25
Part B — Performance measures (all 6, shown)	35
Part C — Little's Law verification	20
Part D — Python confirmation script	20
Total	**100**

Full rubric: `rubrics/homework_rubric.pdf` | *Solutions: the instructor package (available to instructors on request).*